DATA620013 Advanced Statistical Learning

Assignment 1

Exercise 1.

The Gauss–Markov Theorem. Suppose $\hat{\beta}$ is the ordinary least-squares estimator of β in the linear regression model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ ($\mathbf{y}, \varepsilon \in \mathbb{R}^n, \beta \in \mathbb{R}^{p+1}, \mathbf{X} \in \mathbb{R}^{n\times(p+1)}$). Proof that if $\beta^\star$ is any other linear unbiased estimator of β, then $\mathrm{Var}(c^\top \beta^\star) \geq \mathrm{Var}(c^\top \hat{\beta})$, where c is any constant vector of the appropriate order.

Exercise 2.

Theorem of PCA. Suppose x is a m-dim random variable with covariance matrix Σ, $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_m \geq 0$ are the eigenvalues of Σ, and $\alpha_1, \alpha_2, \ldots, \alpha_m$ are the corresponding eigenvectors, then the k-th principle component of x is given by

$$y_k = \alpha_k^\top x = \alpha_{1k}x_1 + \alpha_{2k}x_2 + \ldots + \alpha_{mk}x_m, \tag{1}$$

for $k = 1, 2, \ldots, m$, and the variance of y_k is given by $\mathrm{Var}(y_k) = \alpha_k^\top \Sigma \alpha_k = \lambda_k$ (the k-th eigenvalue of Σ). Please provide a proof of the theorem in the case of the first two principal components (i.e., $k = 1, 2$).

Exercise 3. Programming.

1. Randomly generate 100 datasets, each of which consists of 25 points that are samples of $y = \sin(2\pi x) + \varepsilon$, where $x \in \{0.041 \times i, i = 0, 1, \ldots, 24\}$, and ε is additive

white Gaussian noise with $N(0, 0.3^2)$. Perform ridge regression on each dataset with 7th-order polynomial (with 8 free parameters) with different values of λ. Plot the average results corresponding to different λ (such as $0.001, 0.1, 10, 1000$, etc.), and briefly explain the reason for this result.

2. Implement the iteratively reweighted least squares (IRLS) algorithm for logistic regression models according to textbook section 4.4.1. The attachment data01.txt provides some students' scores on two exams and admission status. Use your own code to train a classifier to determine whether a student is admitted based on the scores of the two exams. Please provide the trained logistic model and its parameters, plot the raw data and decision boundary in one figure, and describe the classification accuracy.

Notes:

1. It's recommended to use LaTeX to complete the assignment, and the document should be submitted in PDF format with the naming format of "Assignment1-张三-23210980001". The assignment template can be found in this document. Please indicate your name, student number and date at the beginning of the document. **Assignments are due in two weeks, so please be aware of the deadlines and submit them on eLearning on time!**

2. The recommended programming language is Python or R with necessary comments. The attached code should be named in the format of "Assignment1-Code-张三-23210980001", or you may combine the code, documentation and results into one.

3. Please complete the assignment independently, discussion is allowed but plagiarism is forbidden. If you have any questions about the assignments, you may contact the TA.